



Network Programming (IE2102)
2nd Year, 1st Semester

Assignment

**Design and Implementation of a Secure Multiprocessor TCP Application
Server with Custom Protocol and Session Management**

Submitted to
Sri Lanka Institute of Information Technology

IT24100626

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In partial fulfillment of the requirements for the
Bachelor of Science Special Honors Degree in Information Technology

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The report must include screenshots of:

a. Hostname configuration

```
gayathri@gayathri-VirtualBox:~/np_assignment$ echo "=== IT24100626 - Hostname ==="
hostnamectl
=== IT24100626 - Hostname ===
  Static hostname: gayathri-VirtualBox
    Icon name: computer-vm
    Chassis: vm
    Machine ID: d8b3b29861114dfef58dcdb786484c5e
    Boot ID: 6dccdd94a43045e0a29d8d70d415cf45
  Virtualization: oracle
  Operating System: Ubuntu 24.04.4 LTS
    Kernel: Linux 6.17.0-19-generic
  Architecture: x86-64
  Hardware Vendor: innotek GmbH
  Hardware Model: VirtualBox
  Firmware Version: VirtualBox
    Firmware Date: Fri 2006-12-01
    Firmware Age: 19y 4month 5d
```

Figure 1

b. Network configuration

```
gayathri@gayathri-VirtualBox:~/np_assignment$ echo "=== IT24100626 - Network Configuration ==="
ip addr show
=== IT24100626 - Network Configuration ===
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c7:ab:d9 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 84383sec preferred_lft 84383sec
    inet6 fd17:625c:f037:2:ad60:3701:81a5:9423/64 scope global temporary dynamic
        valid_lft 86163sec preferred_lft 14163sec
    inet6 fd17:625c:f037:2:a00:27ff:fec7:abd9/64 scope global dynamic mngtmpaddr
        valid_lft 86163sec preferred_lft 14163sec
    inet6 fe80::a00:27ff:fec7:abd9/64 scope link
        valid_lft forever preferred_lft forever
```

Figure 2

c. Server running (port proof using ss/netstat)

```
gayathri@gayathri-VirtualBox:~/np_assignment$ echo "=== IT24100626 - Server Running on Port 50626 ==="
ss -tlnp | grep 50626
=== IT24100626 - Server Running on Port 50626 ===
LISTEN 0      10          0.0.0.0:50626  0.0.0.0:*    users:(("server_0626",pid=4857,fd=4))
```

Figure 3

d. Successful REGISTER and LOGIN

```
gayathri@gayathri-VirtualBox:~/np_assignment$ echo "=== IT24100626 - Successful REGISTER and LOGIN ==="
python3 client_0626.py
=== IT24100626 - Successful REGISTER and LOGIN ===
Connected to server on port 50626

Enter command (or 'quit'): REGISTER Gayathri 0618
Server: OK 200 SID:1006 Registration successful

Enter command (or 'quit'): LOGIN Gayathri 0618
Server: OK 200 SID:1006 Login successful token:TOK-Gayathri-532761
Token saved: TOK-Gayathri-532761

Enter command (or 'quit'): LOGOUT
Server: OK 200 SID:1006 Logout successful

Enter command (or 'quit'): quit
```

Figure 4

e. Failed login + logout proof

```
gayathri@gayathri-VirtualBox:~/np_assignment$ echo "=== IT24100626 - Successful REGISTER and LOGIN ==="
python3 client_0626.py
=== IT24100626 - Successful REGISTER and LOGIN ===
Connected to server on port 50626

Enter command (or 'quit'): LOGIN Gayathri WrongPass1
Server: ERR 401 SID:1006 Invalid credentials

Enter command (or 'quit'): LOGIN Gayathri WrongPass2
Server: ERR 401 SID:1006 Invalid credentials

Enter command (or 'quit'): LOGIN Gayathri WrongPass3
Server: ERR 423 SID:1006 Account locked (brute-force)

Enter command (or 'quit'): LOGIN Gayathri WrongPass4
Server: ERR 423 SID:1006 Account locked (brute-force)
```

Figure 5

f. Multiple forked processes (ps output)

```
gayathri@gayathri-VirtualBox:~/np_assignment$ echo "=== IT24100626 - Multiple Forked Processes (10 Clients) ==="
ps aux | grep server_0626
=== IT24100626 - Multiple Forked Processes (10 Clients) ===
gayathri 4857 0.0 0.0 2928 1772 pts/0 S+ 08:30 0:00 ./server_0626
gayathri 4894 0.0 0.0 2928 1740 pts/0 S+ 08:33 0:00 ./server_0626
gayathri 5445 0.0 0.0 2928 1740 pts/0 S+ 09:07 0:00 ./server_0626
gayathri 5557 0.0 0.0 2928 1736 pts/0 S+ 09:17 0:00 ./server_0626
gayathri 5559 0.0 0.0 2928 1736 pts/0 S+ 09:17 0:00 ./server_0626
gayathri 5564 0.0 0.0 2928 1736 pts/0 S+ 09:17 0:00 ./server_0626
gayathri 5567 0.0 0.0 2928 1736 pts/0 S+ 09:18 0:00 ./server_0626
gayathri 5577 0.0 0.0 2928 1736 pts/0 S+ 09:18 0:00 ./server_0626
gayathri 5590 0.0 0.0 2928 1736 pts/0 S+ 09:19 0:00 ./server_0626
gayathri 5600 0.0 0.0 2928 1736 pts/0 S+ 09:19 0:00 ./server_0626
gayathri 5612 0.0 0.0 2928 1736 pts/0 S+ 09:20 0:00 ./server_0626
gayathri 5633 0.0 0.0 17952 2376 pts/11 S+ 09:21 0:00 grep --color=auto server_0626
```

Figure 6

g. Log file entries

```
gayathri@gayathri-VirtualBox:~/np_assignment$ echo "=== IT24100626 - Server Log File ==="
cat server_IT24100626.log
=== IT24100626 - Server Log File ===
Fri Apr 3 18:30:24 2026 | 127.0.0.1:47006 | 6497 | - | Register | OK 200 SID:1006 Command accepted
Fri Apr 3 18:30:53 2026 | 127.0.0.1:47006 | 6497 | - | REGISTER | OK 200 SID:1006 Command accepted
Fri Apr 3 18:31:22 2026 | 127.0.0.1:47006 | 6497 | testuser | REGISTER | OK 200 SID:1006 Registration successful
Fri Apr 3 18:31:50 2026 | 127.0.0.1:47006 | 6497 | testuser | LOGIN | OK 200 SID:1006 Login successful token:TOK-testuser-221310
Fri Apr 3 18:32:26 2026 | 127.0.0.1:47006 | 6497 | - | LOGOUT | OK 200 SID:1006 Logout successful
Fri Apr 3 18:33:16 2026 | 127.0.0.1:47006 | 6497 | testuser | LOGIN | OK 200 SID:1006 Login successful token:TOK-testuser-221396
Fri Apr 3 18:33:28 2026 | 127.0.0.1:47006 | 6497 | testuser | Hello | OK 200 SID:1006 Command accepted
Fri Apr 3 18:42:05 2026 | 127.0.0.1:54372 | 6620 | amal | REGISTER | OK 200 SID:1006 Registration successful
Fri Apr 3 18:42:36 2026 | 127.0.0.1:54372 | 6620 | amal | LOGIN | OK 200 SID:1006 Login successful token:TOK-amal-221956
Fri Apr 3 18:43:31 2026 | 127.0.0.1:47006 | 6497 | - | LOGOUT | OK 200 SID:1006 Logout successful
Fri Apr 3 18:43:45 2026 | 127.0.0.1:47006 | 6497 | amal | LOGIN | ERR 401 SID:1006 Invalid credentials
Fri Apr 3 18:44:09 2026 | 127.0.0.1:47006 | 6497 | Amal | LOGIN | ERR 401 SID:1006 Invalid credentials
Fri Apr 3 18:44:33 2026 | 127.0.0.1:47006 | 6497 | amal | LOGIN | OK 200 SID:1006 Login successful token:TOK-amal-222073
Fri Apr 3 18:44:43 2026 | 127.0.0.1:47006 | 6497 | - | LOGOUT | OK 200 SID:1006 Logout successful
Fri Apr 3 18:45:13 2026 | 127.0.0.1:47006 | 6497 | amal | LOGIN | ERR 401 SID:1006 Invalid credentials
Fri Apr 3 18:45:24 2026 | 127.0.0.1:47006 | 6497 | amal | LOGIN | ERR 401 SID:1006 Invalid credentials
Fri Apr 3 18:45:32 2026 | 127.0.0.1:47006 | 6497 | amal | LOGIN | ERR 423 SID:1006 Account locked (brute-force)
Fri Apr 3 19:25:28 2026 | 127.0.0.1:35240 | 6733 | amal | LOGIN | OK 200 SID:1006 Login successful token:TOK-amal-224528
Fri Apr 3 19:28:53 2026 | 127.0.0.1:55800 | 7254 | gaya | REGISTER | OK 200 SID:1006 Registration successful
Fri Apr 3 19:29:19 2026 | 127.0.0.1:55800 | 7254 | gaya | LOGIN | OK 200 SID:1006 Login successful token:TOK-gaya-224759
Fri Apr 3 19:30:55 2026 | 127.0.0.1:55800 | 7254 | - | LOGOUT | OK 200 SID:1006 Logout successful
Fri Apr 3 19:33:37 2026 | 127.0.0.1:47006 | 6497 | amal | LOGIN | OK 200 SID:1006 Login successful token:TOK-amal-225017
Fri Apr 3 19:35:23 2026 | 127.0.0.1:55346 | 7346 | gaya | LOGIN | ERR 401 SID:1006 Invalid credentials
Fri Apr 3 19:35:31 2026 | 127.0.0.1:55346 | 7346 | gaya | LOGIN | ERR 401 SID:1006 Invalid credentials
Fri Apr 3 19:35:39 2026 | 127.0.0.1:55346 | 7346 | gaya | LOGIN | ERR 423 SID:1006 Account locked (brute-force)
Fri Apr 3 19:35:56 2026 | 127.0.0.1:55346 | 7346 | gaya | LOGIN | OK 200 SID:1006 Login successful token:TOK-gaya-225156
Fri Apr 3 19:36:36 2026 | 127.0.0.1:55346 | 7346 | gaya | LOGIN | ERR 401 SID:1006 Invalid credentials
Fri Apr 3 19:36:43 2026 | 127.0.0.1:55346 | 7346 | gaya | LOGIN | ERR 401 SID:1006 Invalid credentials
Fri Apr 3 19:36:50 2026 | 127.0.0.1:55346 | 7346 | gaya | LOGIN | ERR 423 SID:1006 Account locked (brute-force)
Fri Apr 3 19:37:01 2026 | 127.0.0.1:55346 | 7346 | gaya | LOGIN | ERR 423 SID:1006 Account locked (brute-force)
Tue Apr 7 08:42:28 2026 | 127.0.0.1:49626 | 4894 | wathi | REGISTER | OK 200 SID:1006 Registration successful
```

Figure 7.1

```
Tue Apr 7 08:42:42 2026 | 127.0.0.1:49626 | 4894 | wathi | LOGIN | OK 200 SID:1006 Login successful token:TOK-wathi-531562
Tue Apr 7 08:42:59 2026 | 127.0.0.1:49626 | 4894 | - | LOGOUT | OK 200 SID:1006 Logout successful
Tue Apr 7 08:43:16 2026 | 127.0.0.1:49626 | 4894 | amal | LOGIN | OK 200 SID:1006 Login successful token:TOK-amal-531596
Tue Apr 7 08:43:27 2026 | 127.0.0.1:49626 | 4894 | - | LOGOUT | OK 200 SID:1006 Logout successful
Tue Apr 7 08:43:38 2026 | 127.0.0.1:49626 | 4894 | wathi | LOGIN | OK 200 SID:1006 Login successful token:TOK-wathi-531618
Tue Apr 7 09:01:43 2026 | 127.0.0.1:46724 | 5355 | Gayathri | LOGIN | ERR 401 SID:1006 Invalid credentials
Tue Apr 7 09:02:27 2026 | 127.0.0.1:34690 | 5384 | Gayathri | REGISTER | OK 200 SID:1006 Registration successful
Tue Apr 7 09:02:41 2026 | 127.0.0.1:34690 | 5384 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-532761
Tue Apr 7 09:06:11 2026 | 127.0.0.1:34690 | 5384 | - | LOGOUT | OK 200 SID:1006 Logout successful
Tue Apr 7 09:07:43 2026 | 127.0.0.1:49930 | 5445 | Gayathri | LOGIN | ERR 401 SID:1006 Invalid credentials
Tue Apr 7 09:08:03 2026 | 127.0.0.1:49930 | 5445 | Gayathri | LOGIN | ERR 401 SID:1006 Invalid credentials
Tue Apr 7 09:08:25 2026 | 127.0.0.1:49930 | 5445 | Gayathri | LOGIN | ERR 423 SID:1006 Account locked (brute-force)
Tue Apr 7 09:13:49 2026 | 127.0.0.1:49930 | 5445 | Gayathri | LOGIN | ERR 423 SID:1006 Account locked (brute-force)
Tue Apr 7 09:16:59 2026 | 127.0.0.1:49626 | 4894 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533619
Tue Apr 7 09:17:22 2026 | 127.0.0.1:34504 | 5557 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533642
Tue Apr 7 09:17:47 2026 | 127.0.0.1:39180 | 5559 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533667
Tue Apr 7 09:18:08 2026 | 127.0.0.1:60078 | 5564 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533688
Tue Apr 7 09:18:24 2026 | 127.0.0.1:49930 | 5445 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533704
Tue Apr 7 09:18:43 2026 | 127.0.0.1:60474 | 5567 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533723
Tue Apr 7 09:19:08 2026 | 127.0.0.1:47538 | 5577 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533748
Tue Apr 7 09:19:30 2026 | 127.0.0.1:56562 | 5590 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533770
Tue Apr 7 09:19:48 2026 | 127.0.0.1:59374 | 5600 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533788
Tue Apr 7 09:20:23 2026 | 127.0.0.1:52610 | 5612 | Gayathri | LOGIN | OK 200 SID:1006 Login successful token:TOK-Gayathri-533823
gayathri@gayathri-VirtualBox:~/np_assignment$
```

Figure 7.2

h. Proper child cleanup (no zombie processes)

```
gayathri@gayathri-VirtualBox:~/np_assignment$ echo "=== IT24100626 - No Zombie Processes ==="
ps aux | grep defunct
=== IT24100626 - No Zombie Processes ===
gayathri  4698  0.0  0.0   0   0 ?        Z    08:29   0:00 [sd_espeak-ng-mb] <defunct>
gayathri  5697  0.0  0.0 17820 2372 pts/11  S+   09:26   0:00 grep --color=auto defunct
```

Figure 8

Important:

Remember

- *Plagiarism is considered as a serious crime against intellectual property in academic environments.*
- *By including the declaration page mentioned below, you abide by the laws of the institute to accept 0 marks for the report, in case you get caught attempting plagiarism.*

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General Formatting

1. Paper size: A4
2. Contents font & size: Times New Roman, size 12
3. Chapter headings/sub headings font & size: Times New Roman Bold, size 12
4. Page numbering:
 - Cover page should not contain any number
 - Declaration page to table of figures page should be numbered using Roman numbers at bottom center of the page
 - Content pages, references page(s) and annexes (if any) should be numbered using Arabic numbers at the bottom center of the page
5. Margins: 1 inch margins on left, right, top and bottom
6. Line spacing: 1 line spacing

Figure Numbering:

1. Each figure should be numbered at bottom center of the figure along with a figure name.
2. Figure number should start with word 'Figure' followed by the 'main chapter number' and 'figure number within the chapter' separated by a 'period sign (.)' and followed by a 'colon (:)'.

For example,

Figure 2.6: XYZ Connector

Declaration Page:

Declaration page should contain the following. Reports that does not contain this declaration 'AS IS' will not be accepted as valid reports and will be awarded 0 marks.

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